Tutorial 7

Question 1: What is a power amplifier? Discuss the differences between a power and a voltage amplifier.

Question 2: Explain the terms related to a power amplifier: Collector efficiency, power dissipation capability, heat sink, Distortion and thermal runaway.

Question 3: Based on the cycle of operation, how can you classify power amplifiers?

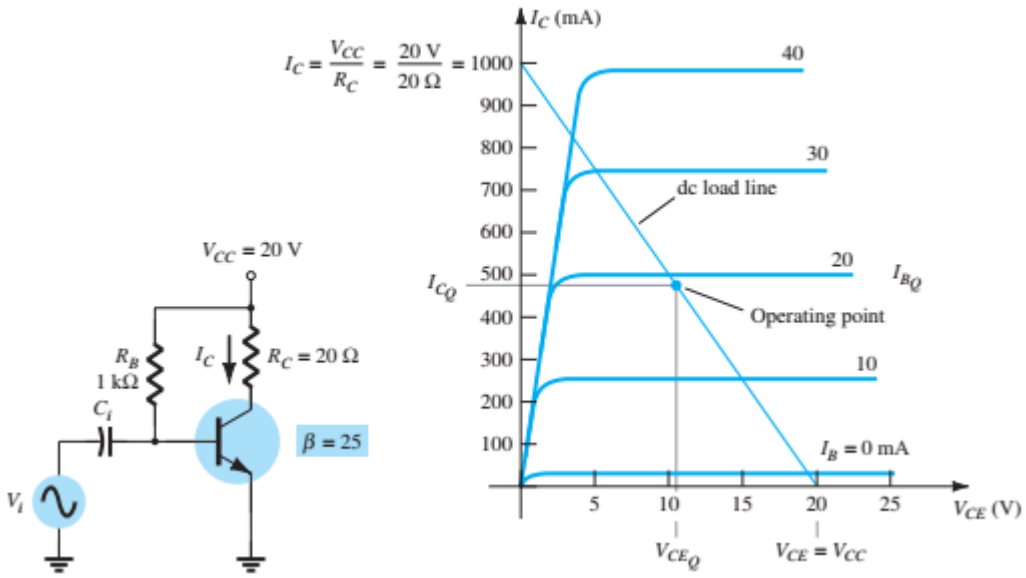
Question 4: What is class A power amplifier? Calculate the maximum efficiency of a Class A power amplifier.

Question 5: What is class B power amplifier? Calculate the maximum efficiency of a Class B power amplifier.

Question 6: What is class AB power amplifier? Explain how a class AB amplifier can remove the crossover distortion?

Question 7: Compare among the class A, Class B, Class AB and class C amplifier.

Problem 1:
Calculate the input power, output power, and efficiency of the amplifier circuit shown below for an input voltage that results in a base current of 10 mA peak.



Problem 2:

Calculate the ac power delivered to the 8-ohm speaker for the circuit below . The circuit component values result in a dc base current of 6 mA, and the input signal (Vi) results in a peak base current swing of 4 mA.

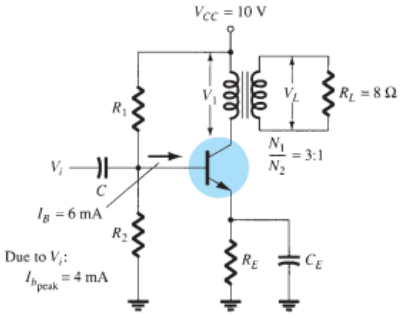
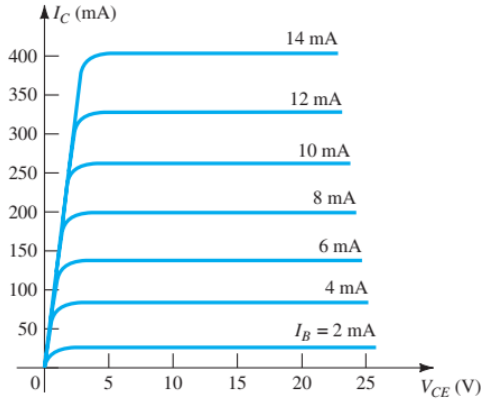


FIG. 12.10

Transformer-coupled class A amplifier for Example 12.4.



Problem 3:

calculate the dc input power, power dissipated by the transistor, and efficiency of the circuit for the input signal results in a peak base current swing of 4 mA

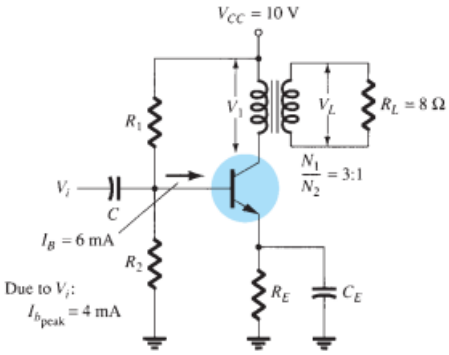
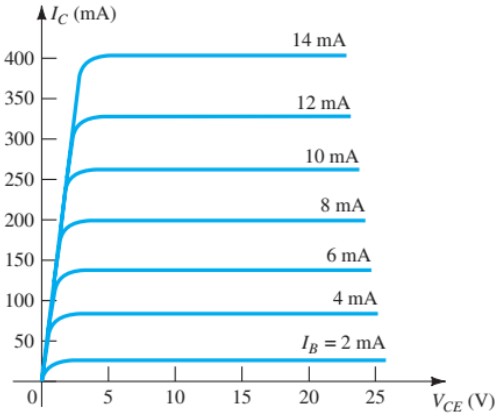


FIG. 12.10

Transformer-coupled class A amplifier for Example 12.4.



Problem 4:

Calculate the efficiency of a transformer-coupled class A amplifier for a supply of 12 V and outputs of:

a. V(p) = 12 V.

b. V(p) = 6 V.

c. V(p)= 2V.

Problem 5:

For a class B amplifier providing a 20-V peak signal to a 16-ohm load (speaker) and a power supply of VCC = 30 V, determine the input power, output power, and circuit efficiency.

Problem 6:

For a class B amplifier using a supply of VCC = 30 V and driving a load of 16 ohm, determine the maximum input power, output power, and transistor dissipation.

Problem 7:

Calculate the efficiency of a class B amplifier for a supply voltage of VCC = 24 V with peak output voltages of:
a. VL(p) = 22 V.
b. VL(p) = 6 V.